Protein-Energy Malnutrition (PEM)

The United Nations estimates that 852 million individuals, including approximately 170 million—about a quarter of the world's children—suffer from protein-energy malnutrition (PEM), with up to 5 million deaths annually attributed to PEM alone.

In developing countries, malnutrition most commonly results from inadequate dietary intake, often exacerbated by disease. Factors such as war, famine, poverty, political unrest, natural disasters, infectious diseases, seasonal and climatic variations, insufficient food production, lack of education, poor sanitation, regional and religious practices, and limited access to health care further restrict food availability. In contrast, in Western industrialized nations like the United States, less than 1% of children suffer from PEM. When it does occur, it is typically associated with chronic illness, malabsorption, presumed food allergies, nutritional ignorance, or adherence to fad diets.

Etiology and Pathogenesis

Protein-energy malnutrition refers to a spectrum of disorders caused by varying degrees of protein and calorie deficiency. Several subtypes of PEM are defined based on the relative deficiencies in protein and total caloric intake.

Marasmus

Children with marasmus experience severe wasting and stunting, with body weight below 60% of the expected weight for age. The term "marasmus" comes from the Greek *marasmos*, meaning "wasting." This condition results from chronic, generalized nutrient deficiency, often due to prolonged food scarcity.

Kwashiorkor

Children with kwashiorkor typically have body weights between 60% and 80% of expected for age. It generally arises from a diet high in carbohydrates—such as grain-based foods—but deficient in protein and fat. This pattern is common in regions where grains are abundant but more expensive proteins and fats are inaccessible. The term "kwashiorkor" originates from a Ghanaian phrase meaning "the one who is deposed," referring to a child weaned from breast milk when a younger sibling is born, transitioning to a carbohydrate-rich, protein-poor diet.

In Western countries, kwashiorkor has been reported in infants fed exclusively on rice-based beverages ("rice milk") as substitutes for infant formula—often due to lower cost or perceived hypoallergenic benefits.

Marasmic Kwashiorkor

A hybrid form of PEM, marasmic kwashiorkor, is characterized by both growth stunting and the presence of edema.

Pathophysiology: Adapted vs. Nonadapted Starvation

The pathophysiology of PEM can be understood in terms of adapted and nonadapted starvation.

Adapted Starvation (Marasmus)

In adapted starvation, chronic deficiency of all macronutrients—especially carbohydrates—leads to reduced insulin secretion. Catabolic hormones dominate, promoting glycogen breakdown into glucose. During the initial 24 hours, muscle protein is broken down to support gluconeogenesis. Subsequently, fat breakdown produces ketone bodies, which the brain and central nervous system can use, reducing the need for further muscle degradation and conserving lean body mass. However, in prolonged starvation, once fat and glycogen stores are depleted, lean tissue is consumed, ultimately leading to death if no nutrients are provided.

Nonadapted Starvation (Kwashiorkor)

In nonadapted starvation, carbohydrate intake is relatively preserved while protein and fat intake are deficient. Insulin secretion continues due to carbohydrate availability, suppressing protein synthesis. This leads to: - **Hypoproteinemia** and subsequent **edema** - **Diarrhea** - Impaired lipoprotein synthesis, resulting in **fatty liver** - Failure to produce essential immune proteins, increasing susceptibility to **opportunistic infections** and **septicemia**, which are leading causes of death in kwashiorkor.

Despite the utility of the adapted vs. nonadapted starvation model, the precise reasons why some children develop marasmus while others develop kwashiorkor under similar conditions of starvation remain unclear and are a subject of ongoing debate.

Clinical Findings

Childhood is a period of rapid growth and development, making it particularly vulnerable to nutritional deficits. Nutritional deprivation in this period often manifests as disruptions in normal growth patterns.

Failure to thrive is a hallmark of PEM and typically presents initially as **wasting**—poor weight gain or weight loss. Over time, it progresses to **stunting**, reflected in slowed linear growth.